

# COMM144 Generative AI 2nd Workshop

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## Exercise 1

Derive the update expression for the E-step when optimizing the lower bound function  $B$  for training a GMM model, which is given by

$$\begin{aligned} B(\{\pi_j, \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j\}_{j=1}^K, \{\gamma_{ij}\}) \\ = \sum_{i=1}^n \sum_{j=1}^K \gamma_{ij} \log \pi_j + \sum_{i=1}^n \sum_{j=1}^K \gamma_{ij} \log \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j) - \sum_{i=1}^n \sum_{j=1}^K \gamma_{ij} \log \gamma_{ij}, \end{aligned}$$
$$\text{subject to } \sum_{j=1}^K \pi_j = 1 \text{ and } \sum_{j=1}^K \gamma_{ij} = 1 \quad \forall i \in \{1, \dots, n\}$$

## Exercise 2

Check Part A, B, C of the python notebook

Part A is for Gaussian distribution

Part B is for GMM

Part C is for Autoencoders.